AI FOR GOOD · HACKATHON 5: MODEL SHOWDOWN

Underpaid and Overworked

Predicting which migrant victims of forced labour had their wages taken, so support workers can help them first

SDG 8.5 Equal pay

SDG 8.7 End forced labour

SDG 8.8 Protect migrant workers

Allan Hassan & Milou van Leuven

scikit-learn · KNN · Logistic Regression · Random Forest

A global problem that hits migrants hardest

27.6M

people in forced labour worldwide

Estimate for 2021

3×

more likely: migrant workers vs. other workers to end up in forced labour

13%

lower pay for migrants than for nationals in high-income countries

Migrant women: about 21% less than male nationals

$236 bn

illegal profits from forced labour every year

Europe & Central Asia: $84bn, the highest of any region

Migrants are over-represented at the bottom of the labour market: paid less, working longer, and most exposed to the worst abuse.

Sources: ILO, Walk Free & IOM (2022), Global Estimates of Modern Slavery · ILO (2020), The Migrant Pay Gap · ILO (2024), Profits and Poverty: The Economics of Forced Labour

In the Netherlands: essential workers, weak protection

~700,000 labour migrants in the Netherlands (2022), a rough underestimate, plus ~140,000 family members

52% of temporary-agency work is done by foreign-born workers (2024); two-thirds of that by people who arrived < 2 years ago

77.5% of EU labour migrants work full-time, compared with 52.3% of all workers

48% of EU labour migrants still earn under 130% of the minimum wage after ten years in the Netherlands

THE VISIBLE TIP

≈230

suspected victims of labour exploitation registered per year (2015–2019). 96% are not Dutch; agriculture is the largest sector (28%).

≈2,500

temp agencies (about 1 in 7) deliberately break the law, says the Labour Inspectorate

Sources: Rijksoverheid (2025), IBO Arbeidsmigratie · CBS (2026), In buitenland geboren uitzendkrachten doen ruim helft uitzendwerk · Nationaal Rapporteur Mensenhandel (2021), Uitgelicht: Arbeidsuitbuiting · Nederlandse Arbeidsinspectie in NRC (2022)

What we mean by the key terms



Labour migrant

A person who moves to another country to work. In our data: a victim whose citizenship differs from the country where the exploitation took place.



Wage exploitation

Being paid less than was earned: withheld or unpaid wages, illegal deductions, or debt bondage (wages used to pay off a debt imposed by the employer or recruiter).



Excessive working hours

Work beyond legal or agreed limits, often as unpaid overtime. One of the eleven ILO indicators of forced labour, together with withholding of wages and debt bondage.



Forced labour

“All work or service which is exacted from any person under the menace of any penalty and for which the said person has not offered himself voluntarily.” (ILO Convention No. 29)

Sources: ILO (1930), Forced Labour Convention No. 29 · ILO (2012), ILO Indicators of Forced Labour

Why migrants end up underpaid and overworked

Dual labour market

Piore (1979)

The labour market splits into a primary segment (secure, well paid) and a secondary segment (insecure, low paid). Migrants fill the secondary jobs locals avoid, with little power to negotiate.

Dependency on one employer

Aanjaagteam (2020)

When one employer or agency arranges the job, the housing and the transport, losing the job means losing the home. Workers stay silent about underpayment.

Continuum of exploitation

Skrivankova (2010)

Exploitation is a scale, not a yes/no: from decent work, through labour-rights violations, to forced labour. Underpayment and long hours are the early steps.

Our dataset covers this end of the scale



Sources: Piore (1979), Birds of Passage · Aanjaagteam Bescherming Arbeidsmigranten (2020), Geen tweederangsburgers · Skrivankova (2010), Between decent work and forced labour, Joseph Rowntree Foundation

SDG 8: Decent work and economic growth

8.5

"…full and productive employment and decent work for all women and men… and equal pay for work of equal value."

In our project: Migrants earn less for the same work.

8.7

"Take immediate and effective measures to eradicate forced labour, end modern slavery and human trafficking…"

In our project: Our target, withheld wages and debt bondage, is a core forced-labour indicator.

8.8

"Protect labour rights and promote safe and secure working environments for all workers, including migrant workers, in particular women migrants…"

In our project: Migrant workers are our population; we check errors for women separately.

One outcome wages withheld / debt bondage · **One group** migrant victims of forced labour · **One decision** who gets wage and debt support first

Source: United Nations (2015), Transforming our world: the 2030 Agenda for Sustainable Development, Goal 8 targets

Who uses it, who it is about, and who is missing

User

Intake caseworker at a victim-support organisation, e.g. CoMensha (national coordination centre against human trafficking) or FairWork

- Meets a newly identified victim of forced labour
- Has limited time and few legal and financial specialists
- Decides who gets priority for a wage claim and debt counselling
- Uses the score as advice; the caseworker decides

People affected

Migrant victims of forced labour

- 9,439 unique cases in our data, 58 nationalities, exploited in 71 countries
- 39% had wages withheld or were in debt bondage
- A missed case means lost wages and a debt that keeps them trapped

Not in the data

People the model knows nothing about

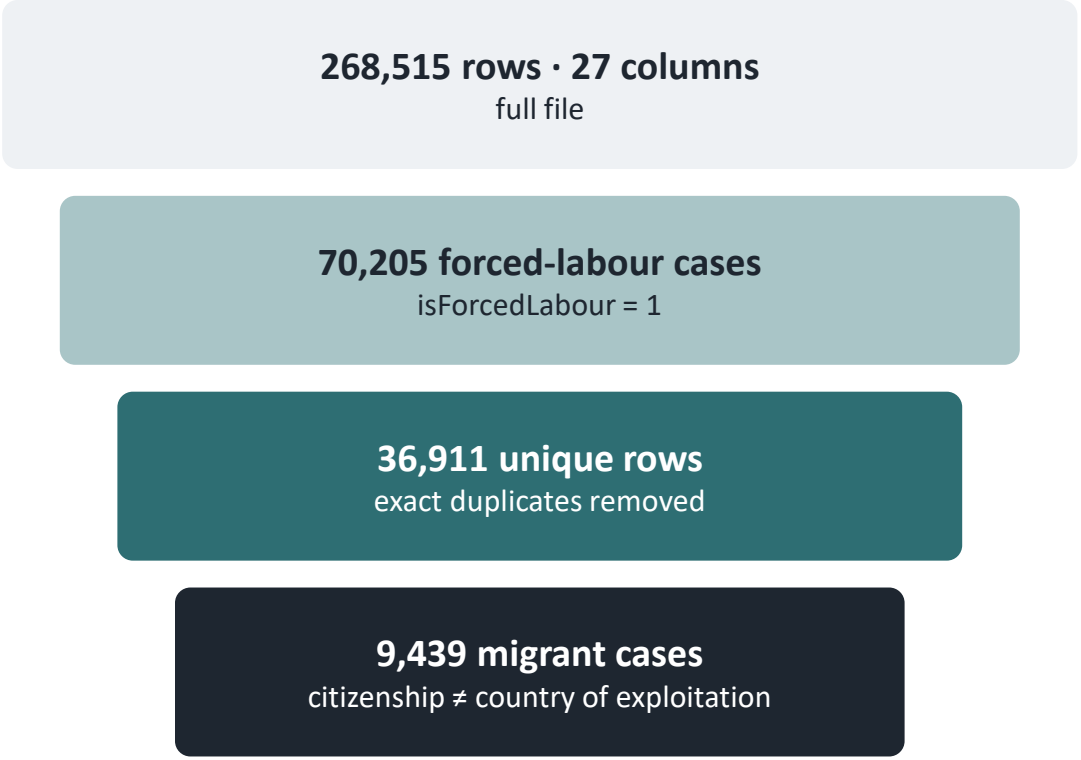
- Victims who were never identified, likely the majority
- Underpaid migrants not (yet) in forced labour
- The Netherlands: fewer than 20 forced-labour cases
- Non-binary people: 5 cases

Not intended for: employers or agencies screening workers · immigration enforcement · judging migrants in general

Sources: CTDC (2026), Global Synthetic Dataset, own analysis · CoMensha, Coördinatiecentrum tegen Mensenhandel · Nationaal Rapporteur Mensenhandel (2021)

The data: CTDC Global Synthetic Dataset

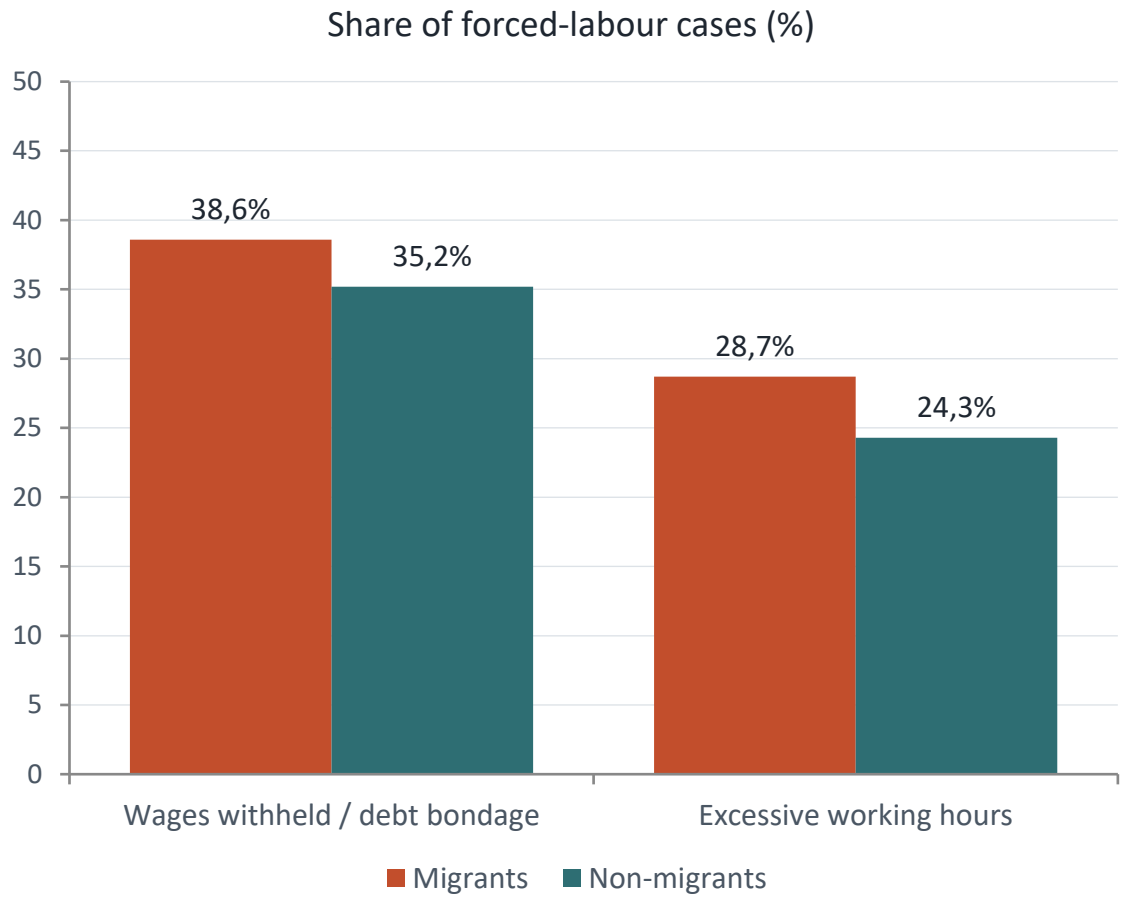
Who	Counter Trafficking Data Collaborative (CTDC), led by IOM, built with Microsoft Research
What	Case records of identified victims of trafficking from IOM, Polaris, A21 and other organisations
Scale	Based on 206,000+ real victims from 190 countries; registration years 2002–2024
How	Synthetic rows generated with differential privacy: same statistics, no real person can be traced
Access	Free public download with codebook at ctdatacollaborative.org

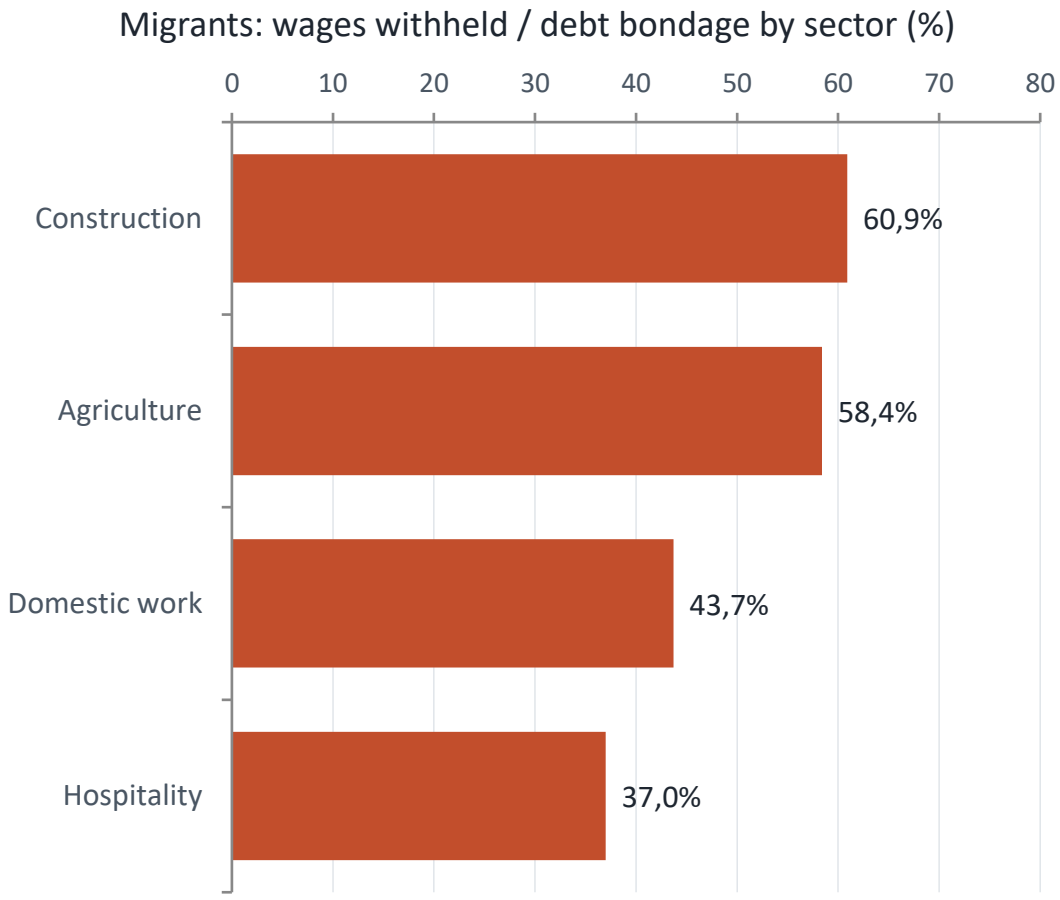


Target: 38.6% had wages withheld or were in debt bondage

Sources: IOM (2024), IOM releases the Global Synthetic Dataset · CTDC (2026), ctdc_global_synthetic_data_v2026.csv, own analysis

Migrant victims lose wages and work too long more often





Source: CTDC (2026), own analysis after removing duplicates: 9,439 migrant vs 2,051 non-migrant forced-labour cases. A blank field means 'not recorded', so these are minimum shares.

The prediction: did this person lose their wages?

Target (y)

isLaborForced

Features (X)

Gender, age band, citizenship, country of exploitation, sector, relationship to recruiter, year of registration, duration

Left out

All columns that starts with "mean..." are dropped, beacause this columns were filled after an inspection was done in real life.

False negative: the worse mistake
The model says ‘not forced’ but the worker was actually forced.

False positive
The model flags wage a forced labor that did not happen. Cost: one extra financial check by the caseworker.

Main metric: recall
Find as many victims with lost laborforced as possible. F1 is reported alongside, so a model cannot ‘win’ by flagging everyone.

From raw file to recommendation: our pipeline

1 Load

Raw CSV in, with pandas

2 Split

80/20, stratified, random_state = 42

3 Clean

Dropped unwanted columns cleaned data.

scikit-learn

4 Baseline

Created DummyClassifier - most frequent class

scikit-learn

5 Train

Trained 3 models:
KNN
Logistic Reg
Random Forest

scikit-learn

6 Tune

GridSearchCV, same 5 stratified folds, scoring = recall

scikit-learn

7 Test once

Confusion matrix, precision, recall, F1

scikit-learn

8 Evaluate

Display test result of all models

scikit-learn

9 Use

Recommend a model, predict a new case with probability

scikit-learn

Models tuned: KNN (n_neighbors) · Logistic Regression (C) · Random Forest (max_depth, n_estimators)

Model showdown: results

Model	Test precision	Test recall	Missed victims	Caught victims
Baseline (DummyClassifier)	0.00%	0.00%	14,041	0
KNN	59.65%	50.47%	6,955	7,086
Logistic Regression	68.20%	39.97%	8,429	5,612
Random Forest	70.47%	50.44%	6,959	7,082

How to read this: fill in the numbers from the notebook's comparison table. A model only counts as better when the difference is larger than the spread over the folds. Train, CV and test recall side by side show over- or underfitting.

Which model we recommend?

- Random Forest
- KNN caught 4 more victims (7,086 vs 7,082), KNN triggered **4,794** false alarms.
- Random Forest caught almost the same number of victims while cutting false alarms down to **2,967**. This mean a precision of (70.47%) for Random forest.
- This ensures inspectors focus their time and resources on real risk cases rather than wasting effort on safe businesses.

Ethical risks and what we did about them

Risk	Consequence for the people in the data	What we did / will do
Synthetic data: rows are not real people; victims were not asked	Patterns learned from synthetic rows may not hold for real victims	Prototype only. README warns: retrain on real case data under a data agreement before any use
Citizenship as a feature can turn into profiling	Migrants could be flagged or overlooked because of their passport; misuse by enforcement	Use limited to victim support; ‘not intended for’ list in README; next step: compare with and without citizenship
Blank means ‘not recorded’, treated as ‘no’	Poorly documented cases look safe: wage loss is 50% for men but 26% when gender is unknown → more false negatives	Recall checked per gender group; threshold can be lowered to catch more cases
Population mismatch: mostly USA and Russia, fewer than 20 Dutch cases	Results may not hold for Polish or Romanian workers in Dutch agriculture	Stated as a limitation; not for direct use in the Netherlands

Sources: CTDC (2026), own analysis · IOM (2024) on differential privacy

Sources

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